



Manish Singh

Master of Technology
in Computer Science Engineering
Indian Institute Of Technology, Ropar

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EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Master of Technology	Indian Institute of Technology, Ropar	8.3	2025-2027
Bachelor of Technology	Haldia Institute of Technology, West Bengal	9.06	2017-2021
Senior Secondary	Central Board of Secondary Education	91.6%	2016
Secondary	Central Board of Secondary Education	10	2014

EXPERIENCE

- Intel** Jul 2025 - Present
AI Software Engineer Graduate Intern Bengaluru
 - Building agentic AI frameworks to automate and optimize AI workload development, integrating tool-calling and multi-step task orchestration across software and infrastructure layers.
 - Collaborating on the design of autonomous AI agent architectures to support performance profiling and system-level workflow automation, aligned with Intel's AI SoC development goals.
- Infosys** Aug 2021 - Oct 2024
Specialist Programmer Pune
 - Engineered Spark analytical pipelines in Azure HDInsight, orchestrating data processing through Azure Data Factory (ADF); extracted actionable insights from datasets, reducing processing time by 30% and boosting data reliability.
 - Optimized pipeline performance, resulting in a notable 25% reduction in total processing time.
 - Developed robust CI/CD pipelines using Azure DevOps; reduced deployment time by 40% and increased release frequency from monthly to bi-weekly, enhancing overall efficiency.
 - Technologies: Azure HDInsight, Azure DataFactory, Azure DevOps, Spark, Scala, Azure SQL Database

PROJECTS

- State-wise YouTube Trend and Sentiment Analysis using NLP and BDA** Jan 2026 - May 2026
Course GitHub
 - Developed a system to analyze YouTube trends and sentiment at a state level by extracting topics, mapping them to regions, and visualizing insights via a dashboard.
 - Built the NLP pipeline including multilingual preprocessing, semantic topic modeling (Sentence-BERT + KeyBERT), sentiment analysis (VADER), and integration with PySpark for scalable processing.
 - Tech Stack: Python, PySpark, Sentence-BERT, KeyBERT, VADER, FLAN-T5, Delta Lake, Streamlit, Docker
- Music Genre Classification on GTZAN** Jan 2026 - May 2026
Course GitHub
 - Built an end-to-end music genre classification pipeline on the GTZAN dataset, comparing multiple machine learning and deep learning approaches for robust genre prediction.
 - Performed feature engineering, model training, and evaluation to analyze performance trade-offs and improve classification reliability for AI-focused use cases.
 - Tech Stack: Python, PyTorch, Jupyter Notebook, NumPy, Pandas, Scikit-learn, Librosa, Matplotlib, Seaborn
- Campus Court - Sports Facility Management App** July 2025 - Nov 2025
Course GitHub
 - Engineered a full-stack Android application for real-time booking of campus sports facilities and student team management at IIT Ropar.
 - Implemented secure authentication and live slot availability using cloud-hosted APIs, supporting concurrent users and conflict-free bookings.
 - Tech Stack: Django (REST Framework), Redis (Caching), Supabase (PostgreSQL), Flutter, Render (Cloud Hosting)

TECHNICAL SKILLS

- Programming Languages:** C/C++, Python, SQL
- Machine Learning & AI:** Machine Learning, Deep Learning, Computer Vision, PyTorch, LangGraph, LangChain, RAG, Pinecone
- Backend & Databases:** Django, PostgreSQL, MySQL
- Big Data & Cloud:** Spark, Spark SQL, PySpark, Kafka, Spark Streaming, Microsoft Azure, Azure Data Factory, Databricks
- DevOps & Tools:** Docker

KEY COURSES TAKEN

- CSE & Maths:** Data Structure & Algorithm, Operating System, DBMS, Computer Systems, Mathematical Foundation for Computer Science, Internet of Things (IoT), Machine Learning, Deep Learning for Computer Vision

ACHIEVEMENTS

- GATE Score CSE: 739 2025
- Received outstanding performance rating for exemplary project contributions and dedication 2022
- INSTA Award, Certificate of Appreciation for Excellent Performance in a project from Infosys 2022